FORMULE di GREEN-GAUSS

(SERVONO PER TRASFORMARE INTEGRALI DOPPI IN CURVILINEI E VICEVERSA)

[a] SIA $f = f(x,y)$ CONTINUA SU UN DOMINIO $\mathcal{D} \subseteq \mathbb{R}^2$ NORMALE RISPETTO ALL'ASSE y E SIA ANCHE f_x CONTINUA IN $\mathcal{D}$.

ALLORA $\iint_{\mathcal{D}} \frac{\partial F}{\partial x}\,dx\,dy = \int_{+F\mathcal{D}} f\,dy$ ← $\left(\omega = 0\,dx + f\,dy \;\; \int_{+F\mathcal{D}} \omega\right)$

INTEGRALE DOPPIO SU $\mathcal{D}$ — INTEGRALE CURVILINEO SU $+F\mathcal{D}$

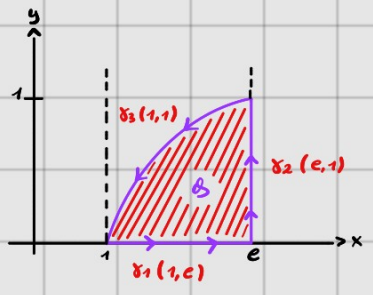
[b] SIA $f = f(x,y)$ CONTINUA SU $\mathcal{D} \subseteq \mathbb{R}^2$ NORMALE RISPETTO ALL'ASSE x. SIA ANCHE f_y CONTINUA IN $\mathcal{D}$.

ALLORA $\iint_{\mathcal{D}} \frac{\partial F}{\partial x}\,dx\,dy = -\int_{+F\mathcal{D}} f\,dy$ ← $\left(\omega = f\,dx + 0\,dy \;\; \int_{+F\mathcal{D}} \omega\right)$

INTEGRALE DOPPIO SU $\mathcal{D}$ — INTEGRALE CURVILINEO SU $+F\mathcal{D}$

ESERCIZIO

$\iint_{\mathcal{D}} \frac{y}{x}\,dx\,dy$ $\quad \mathcal{D} = \{1 \le x \le e,\ 0 \le y \le \ln x\}$ NORMALE RISPETTO ASSE X



- VOGLIO APPLICARE [b]

$\iint_{\mathcal{D}} \frac{y}{x}\,dx\,dy = \iint_{\mathcal{D}} \frac{\partial f}{\partial y}\,dx\,dy$

$\frac{\partial f}{\partial y} = \frac{y}{x} \Rightarrow f = \frac{y^2}{2x}$ (HO INTEGRATO $f_y = \frac{y}{x}$ RISPETTO A y) $\Rightarrow \iint_{\mathcal{D}} \frac{y}{x}\,dx\,dy \overset{[b]}{=} -\int_{+F\mathcal{D}} \frac{y^2}{2x}\,dx$

$+F\mathcal{D} = \gamma_1 - \gamma_2 - \gamma_3$ (PERCHÈ γ_3 HA ORIENTAZIONE NEGATIVA)

$-\int_{+F\mathcal{D}} \frac{y^2}{2x}\,dx = -\int_{\gamma_1} - \int_{\gamma_2} + \int_{\gamma_3} = -0 - 0 + \frac{1}{6} = \frac{1}{6}$

γ_1 VA DA $(1,0) \longrightarrow (e,0)$

$\gamma_1 : \begin{cases} x = t \\ y = 0 \end{cases}$ $t \in [1,e]$ — t VARIA IN x; 0 NON VARIA IN y

$dx = dt \quad dy = 0$

$\int_{\gamma_1} \frac{y^2}{2x}\,dx = \int_1^e \frac{0^2}{2t}\,dt = \int_1^e 0\,dt = 0$

γ_2 VA DA $(e,0) \longrightarrow (e,1)$

$\gamma_2 : \begin{cases} x = e \\ y = t \end{cases}$ $t \in [0,1]$

$dx = 0 \quad dy = dt$

$\int_{\gamma_2} \frac{y^2}{2x}\,dx = \int_0^1 \frac{t^2}{2e}\,dx = \int_0^1 \frac{t^2}{2e} \cdot \overset{dx=0}{0} = 0$

γ_3 VA DA $(e,1) \longrightarrow (1,0)$

$\gamma_3 : \begin{cases} x = t \\ y = \ln t \end{cases}$ $t \in [1,e]$

$dx = 1 \quad dy = \frac{1}{t}\,dt$

$\int_{\gamma_3} \frac{y^2}{2x}\,dx = \int_1^e \frac{\ln^2 t}{2t}\,dt = \int_0^1 \frac{v^2}{2}\,dv = \frac{1}{2}\left[\frac{v^3}{3}\right]_0^1 = \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6}$

SOSTITUZIONE: $\ln t = v$, $\frac{1}{t}\,dt = dv$ $\quad t \in [1,e] \longrightarrow v \in [0,1]$

PER SICUREZZA CALCOLIAMO ANCHE L'INTEGRALE DOPPIO:

$\iint_{\mathcal{D}} \frac{y}{x}\,dx\,dy \quad \mathcal{D} = \{1 \le x \le e,\ 0 \le y \le \ln x\}$

$\iint_{\mathcal{D}} \frac{y}{x}\,dx\,dy = \int_1^e dx \int_0^{\ln x} \frac{y}{x}\,dy = \int_1^e \frac{1}{x}\,dx \int_0^{\ln x} y\,dy = \int_1^e \frac{1}{x}\left[\frac{y^2}{2}\right]_0^{\ln x} dx = \int_1^e \frac{1}{x}\left[\frac{\ln^2 x}{2} - \frac{0^2}{2}\right] dx = \int_1^e \frac{\ln^2 x}{2x}\,dx = \frac{1}{6}$

$\int \frac{\ln^2 x}{2x}\,dx = \int \frac{t^2}{2}\,dt = \frac{1}{2}\int t^2\,dt = \frac{1}{2}\cdot\frac{t^3}{3} = \frac{1}{2}\cdot\frac{\ln^3(x)}{3} = \frac{\ln^3(x)}{6}\Big|_1^e = \frac{\ln^3(e)}{6} - \frac{\ln^3(1)}{6} = \frac{1}{6}$

$\ln(e) = 1 \quad \ln(1) = 0$

ESERCIZIO

$\int_{+F\mathcal{D}} 3x^2 y\,dy \overset{[a]}{=} \iint_{\mathcal{D}} 6xy\,dx\,dy$

DOVE $F\mathcal{D}$ È LA FRONTIERA DEL DOMINIO COMPRESO TRA LA CURVA DI EQUAZIONI PARAMETRICHE $x = \frac{1}{2}t - t^2$, $y = t^2$, $0 \le t \le \frac{1}{2}$ E L'ASSE y



$\mathcal{D}\ \{0 \le y \le \frac{1}{4},\ 0 \le x \le \frac{1}{2}\sqrt{y} - y\}$

$y = t^2$
$0 \le t \le \frac{1}{2}$
$0 \le \sqrt{y} \le \frac{1}{2}$
$0^2 \le (\sqrt{y})^2 \le (\frac{1}{2})^2$
$0 \le y \le \frac{1}{4}$

$x = \frac{1}{2}t - t^2$
$t^2 = y \rightarrow t = \sqrt{y}$
$\sqrt{t^2} = \sqrt{y} \longrightarrow t = \sqrt{y}$
$x = \frac{1}{2}\sqrt{y} - y$
$0 \le x \le \frac{1}{2}\sqrt{y} - y$

γ_1 VA DA $(0,0) \longrightarrow (0,\frac{1}{4})$

$\gamma_1 : \begin{cases} x = \frac{1}{2}t - t^2 \\ y = t^2 \end{cases}$ $t \in [0, \frac{1}{2}]$

γ_2 VA DA $(0,\frac{1}{4}) \longrightarrow (0,0)$

$-\gamma_2 : \begin{cases} x = 0 \\ y = t \end{cases}$ $t \in$

$\int_{+FA} 3x^2 y\,dy \overset{[b]}{=} \iint_D 6xy\,dx\,dy = \int_0^{\frac{1}{4}} y\,dy \int_0^{\sqrt{y}} 6x\,dx = \int_0^{\frac{1}{4}} y\,dy \left[\frac{x^2}{2}\right]_0^{\sqrt{y}} = \int_0^{\frac{1}{4}} 3y\left(\sqrt{y}-y\right)^2 dy = \int_0^{\frac{1}{4}} 3y\left(\frac{1}{2}y + y^2 - y^{\frac{3}{2}}\right)dx =$

$= \int_0^{\frac{1}{4}} \frac{3}{4}y^2 + 3y^3 - 3y^{\frac{5}{2}}\,dy = \left[\frac{1}{4}\cdot\frac{y^3}{3} + 3\frac{y^4}{4} - 3\frac{y^{\frac{7}{2}}}{\frac{7}{2}}\right]_0^{\frac{1}{4}} = \left[\frac{1}{4^4} - \frac{3}{4^3} - \frac{6}{7}\cdot\frac{1}{2^7}\right]$

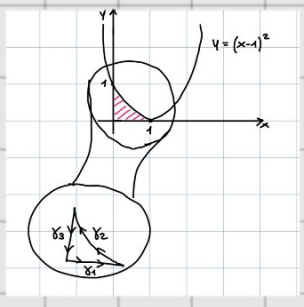
$\int y^n dy = \frac{y^{n+1}}{n+1}$

$\frac{3}{4}y^2 = \frac{1}{4}\cdot\frac{y^3}{3} = \frac{y^3}{4}$ $\qquad 3y^3 = 3\cdot\frac{y^4}{4} = \frac{3}{4}y^4$

$-3y^{\frac{5}{2}} = -3\cdot\frac{y^{\frac{5}{2}+1}}{\frac{5}{2}+1} = -3\cdot\frac{y^{\frac{7}{2}}}{\frac{7}{2}} = -\frac{6}{7}y^{\frac{7}{2}}$

ESERCIZIO

$\iint_A xy\,dx\,dy \qquad A = \{0 \le x \le 1 \quad 0 \le y \le (x-1)^2\}$



DOVE A È:
IL DOMINIO DELIMITATO DALLA CURVA $y=(x-1)^2$ CON $x \in [0,1]$ E DAGLI ASSI NEL PRIMO QUADRANTE

NORM. RISP. ASSE x

$\iint_A xy\,dx\,dy \overset{[b]}{=} -\int_{+FA} x\frac{y^2}{2}\,dx$

PARAMETRIZZIAMO +FA, OSSIA LE CURVE $\gamma_1, \gamma_2, \gamma_3$

γ_1 VA DA $(0,0) \to (1,0)$

$\gamma_1: \begin{cases} x = t \\ y = 0 \end{cases} \quad t \in [0,1]$

$dx = dt \quad dy = 0$

γ_2 VA DA $(1,0) \to (0,1)$

$-\gamma_2: \begin{cases} x = t \\ y = (t-1)^2 \end{cases} \quad t \in [0,1]$

$dx = dt \quad dy = 2(t-1)\cdot 1\,dt$

γ_3 VA DA $(0,1) \to (0,0)$

$\gamma_3: \begin{cases} x = 0 \\ y = t \end{cases} \quad t \in [0,1]$

$dx = 0 \quad dy = dt$

SOSTITUZIONE
$t-1 = v \to t = v+1$
$dt = dv$
$t \in [0,1] \to v \in [-1,0]$

$\iint_A xy\,dx\,dy \overset{[b]}{=} -\int_{+FA} x\frac{y^2}{2}\,dx = -\int_{\gamma_1} + \int_{\gamma_2} + \int_{\gamma_3} = -\int_0^1 t\cdot\frac{0^2}{2}\,dt + \int_0^1 \frac{t\cdot(t-1)^4}{2}\,dt + \int_0^1 0\cdot\frac{t^2}{2}\,0 = \int_0^1 \frac{t\cdot(t-1)^4}{2}\,dt = \frac{1}{2}\int_0^1 t\cdot(t-1)^4\,dt = \int_{-1}^0 (v+1)\cdot v^4\,dv =$

$= \frac{1}{2}\left[v^5 + v^4\,dv\right]_{-1}^0 = \frac{1}{2}\cdot\left[\frac{v^6}{6} + \frac{v^5}{5}\right]_{-1}^0 = \frac{1}{2}\cdot\left[0 - \left(\frac{1}{6} - \frac{1}{5}\right)\right] = \frac{1}{2}\cdot\left[-\frac{1}{6} + \frac{1}{5}\right] = \frac{1}{2}\cdot\frac{1}{30} = \frac{1}{60}$

CALCOLARE L'INTEGRALE DOPPIO PER VERIFICARE IL RISULTATO

SOSTITUZIONE
$x-1 = v \to x = v+1$
$dt = dv$
$t \in [0,1] \to v \in [-1,0]$

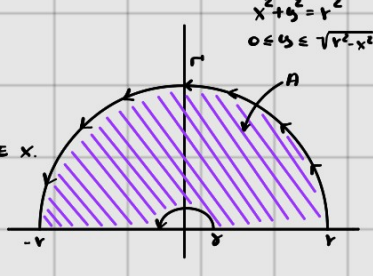
$\iint_A xy\,dx\,dy = \int_0^1 dx \int_0^{(x-1)^2} xy\,dy = \int_0^1 x\,dx \int_0^{(x-1)^2} y\,dy = \int_0^1 x\,dx\cdot\left[\frac{y^2}{2}\right]_0^{(x-1)^2} = \int_0^1 x\,dx\cdot\frac{((x-1)^2)^2}{2} = \int_0^1 x\cdot\frac{(x-1)^4}{2}\,dx = \frac{1}{2}\int_0^1 x(x-1)^4\,dx = \frac{1}{2}\int_{-1}^0 (v+1)\,v^4\,dv = \frac{1}{2}\int_{-1}^0 v^5 + v^4\,dv =$

$= \frac{1}{2}\left[\frac{v^6}{6} + \frac{v^5}{5}\right]_{-1}^0 = \frac{1}{2}\left[0 - \left(\frac{1}{6} - \frac{1}{5}\right)\right] = \frac{1}{2}\left[\frac{1}{30}\right] = \frac{1}{60}$

ESERCIZIO

"CONTINUO ESERCIZIO PRECEDENTE"

$x^2+y^2 = r^2$
$0 \le y \le \sqrt{r^2-x^2}$



IL SEMICERCHIO LIMITATO DALLA CURVA $x = r\cos t$, $y = r\sin t$, CON $r > 0$, $t \in [0,\pi]$ E DALL'ASSE x.

$\iint_A xy\,dx\,dy \qquad \Gamma: \begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \quad t \in [0,\pi]$ $(r>0$ COSTANTE$)$

$+FA = \Gamma + \gamma$

[a] $\iint_D \frac{\partial f}{\partial x}\,dx\,dy = \int_{+FD} f\,dy$

[b] $\iint_D \frac{\partial g}{\partial y}\,dx\,dy = -\int_{+FD} g\,dy$

Γ VA DA $(r,0) \to (-r,0)$

$\gamma: \begin{cases} x = t \\ y = 0 \end{cases} \quad t \in (-r,r)$

γ VA DA $(-r,0) \to (r,0)$

[a] $\iint_D \frac{\partial f}{\partial x}\,dx\,dy = \int_{+FD} f\,dy$

$\iint_A xy\,dx\,dy = \int_{+FA} \frac{x^2y}{2}\,dy = \int_\Gamma + \int_\gamma = \int_\Gamma \frac{x^2y}{2}\,dy + \int_\gamma \frac{x^2y}{2}\,dy =$

$= \int_0^\pi \frac{(r\cos t)^2\cdot r\sin t}{2}\cdot r\cos t\,dy + \int_{-r}^r \frac{t^2\cdot 0}{2}\cdot 0\,dy =$

$= \frac{r^4}{2}\int_0^\pi \cos^3 t\sin t\,dt = \frac{r^4}{2}\left[-\frac{\cos^4 t}{4}\right]_0^\pi = \frac{r^4}{2}\left[-\frac{\cos^4\pi}{4} - 0\right] = 0$

[b] $\iint_D \frac{\partial g}{\partial y}\,dx\,dy = -\int_{+FD} g\,dy$

$\iint_A xy\,dx\,dy = -\int_{+FA} \frac{xy^2}{2}\,dx = -\int_\Gamma \frac{x^2y}{2}\,dx - \int_\gamma \frac{x^2y}{2}\,dx =$

$= -\int_0^\pi \frac{r\cos t\cdot(r\sin t)^2}{2}\cdot(-r\sin t\,dt) - \int_{-r}^r \frac{t\cdot 0^2}{2}\cdot 0\,dt =$

$= \frac{r^4}{2}\int_0^\pi \cos t\cdot\sin^3 t\,dt - 0 = \frac{r^4}{2}\left[\frac{\sin^4 t}{4}\right]_0^\pi = 0$

$\omega = 0\,dx + \frac{x^2y}{2}\,dy \longrightarrow$ NON È UNA FORMA ESATTA, MA $\oint_{+FA} \omega = 0$

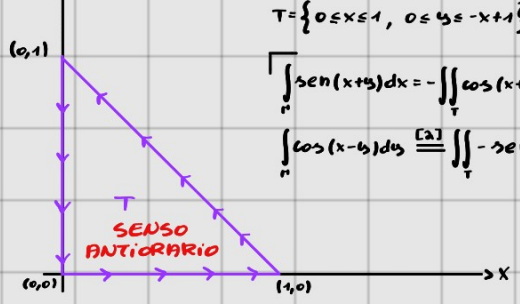


ESERCIZIO

SI CALCOLI L'INTEGRALE DELLA FORMA DIFFERENZIALE

$\omega = \sin(y+x)\,dx + \cos(x-y)\,dy$ LUNGO LA CURVA DATA

DAI LATI DEL TRIANGOLO DEI VERTICI $(0,0)$, $(1,0)$ E $(0,1)$



$T = \{0 \le x \le 1,\ 0 \le y \le -x+1\}$

$\int_\Gamma \sin(x+y)\,dx = -\iint_T \cos(x+y)\,dx\,dy$

$\int_\Gamma \cos(x-y)\,dy \overset{[2]}{=} \iint_T -\sin(x-y)\,dx\,dy$

$$\int_\Gamma \omega = ? = \int_\Gamma \sin(y+x)\,dx + \cos(x-y)\,dy = \int_\Gamma \sin(x+y)\,dx + \int \cos(x-y)\,dy = -\iint_T \cos(x+y)\,dx\,dy - \iint_T \sin(x-y)\,dx\,dy =$$

$$= -\iint_T \cos(x+y) + \sin(x-y)\,dx\,dy = -\int_0^1 dx \int_0^{-x+1} \cos(x+y) + \sin(x-y)\,dy = -\int_0^1 \left[\sin(x+y) + \cos(x-y)\right]_0^{-x+1} dx =$$

$$= -\int_0^1 \left[\left(\sin(x-x+1) + \cos(x-(-x+1)\right) - \left(\sin(x+0) + \cos(x-0)\right)\right] dx = -\int_0^1 \left[\sin 1 + \cos(2x-1) - \sin x - \cos x\right] dx =$$

$$= -\left[(\sin 1)\cdot x + \frac{\sin(2x-1)}{2} + \cos x - \sin x\right]_0^1 = -\left[\left(\sin 1 + \frac{\sin 1}{2} + \cos 1 - \sin 1\right) - \left((\sin 1)\cdot 0 + \frac{\sin(-1)}{2} + \cos(0) - \sin 0\right)\right] =$$

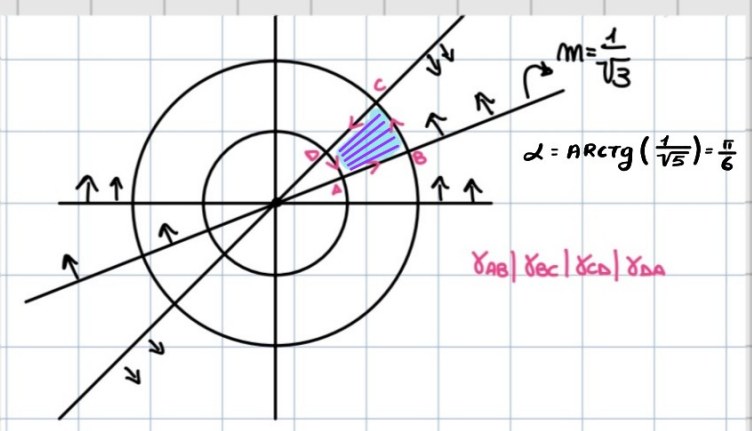
$$= -\left[\frac{\sin 1}{2} + \cos 1 \underbrace{- \frac{\sin(-1)}{2}}_{+\frac{\sin 1}{2}} - 1\right] = -\sin 1 - \cos 1 + 1$$

ESERCIZIO

$$\iint_\Omega \frac{y}{\sqrt{x^2+y^2}}\,dx\,dy \qquad \Omega: \left\{(x,y) \in \mathbb{R}^2 \mid 4 \le x^2+y^2 \le 16 \quad 0 \le \frac{x}{\sqrt{3}} \le y \le x\right\}$$

$\frac{x}{\sqrt{3}} \ge 0 \quad x \ge 0 \qquad y \ge \frac{x}{\sqrt{3}}x$

x	y
0	0
$\sqrt{3}$	1



$\oint f\,dx + g\,dy = \iint \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\,dx\,dy$

PARAMETRIZZAZIONE

$\Omega: \begin{cases} x = \rho\cos\theta & \rho \in [2,4] \\ y = \rho\sin\theta & \theta \in [\frac{\pi}{6}, \frac{\pi}{4}] \end{cases}$

COME INTEGRALE DOPPIO:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} d\theta \int_2^4 \frac{\rho\sin\theta}{\sqrt{\rho^2\cos^2\theta + \rho^2\sin^2\theta}}\,\rho\,d\rho = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} d\theta \int_2^4 \frac{\rho\sin\theta}{\rho}\,\rho\,d\rho = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} d\theta \int_2^4 \rho\sin\theta\,d\rho = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin\theta\,d\theta \int_2^4 \rho\,d\rho = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin\theta\,d\theta \left[\frac{\rho^2}{2}\right]_2^4 = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin\theta\,d\theta \left(\frac{16}{2} - \frac{4}{2}\right) =$$

$$= 6\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin\theta\,d\theta = 6\left[-\cos\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = 6\left[-\cos\frac{\pi}{6} + \cos\frac{\pi}{4}\right] = 6\left[\frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2}\right]$$

ESERCIZIO

CALCOLARE $\int_{Fr\Omega^+} 2(x^2-1)y^2\,dx$ DOVE $Fr\Omega$ È LA FRONTIERA DEL DOMINIO $\Omega = \left\{0 \le x \le 2,\ x-2 \le y \le \sqrt{2x-x^2}\right\}$

[a] $\iint_\Omega \frac{\partial f}{\partial x}\,dx\,dy = \int_{+Fr\Omega} f\,dy$

[b] $\iint_\Omega \frac{\partial g}{\partial y}\,dx\,dy = -\int_{+Fr\Omega} g\,dy$

$$\int_{+Fr\Omega} 2(x^2-1)y^2\,dx = \iint_\Omega 2(x^2-1)2y\,dx\,dy = 2\iint_\Omega (x^2-1)2y\,dx\,dy = 2\int_0^2 (x^2-1)\,dx \int_{x-2}^{\sqrt{2x-x^2}} 2y\,dy = 2\int_0^2 (x^2-1)\,dx \left[y^2\right]_{x-2}^{\sqrt{2x-x^2}} dx = 2\int_0^2 (x^2-1)\left[2x-x^2-(x-2)^2\right] dx =$$

$$= 2\int_0^2 2x^3 - x^4 - x^2(x-2)^2 - 2x + x^2 + (x-2)^2\,dx = 2\int_0^2 2x^3 - x^4 - x(x-2)^2 - 2x + x^2 + x^2 - 4x + 4\,dx = 2\int_0^2 2x^3 - x^4 - x^4 + 4x^3 - 4x^2 - 2x + x^2 + x^2 - 4x + 4\,dx =$$

$$= 2\int_0^2 6x^3 - 2x^4 - 2x^2 - 6x + 4\,dx = 2\left[\frac{3x^4}{2} - \frac{2x^5}{5} - \frac{2x^3}{3} - 3x^2 + 4x\right]_0^2 = 2\cdot\frac{28}{15} = -\frac{56}{15}$$